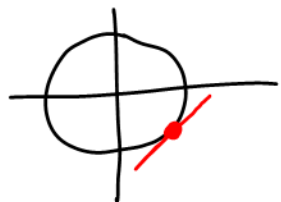


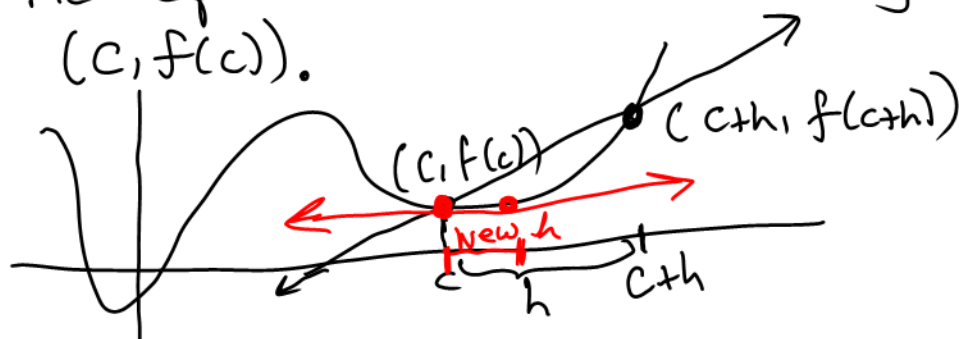
Lecture 5: The Derivative

GOAL: Become familiar with the limit definition of the derivative. Be able to solve the Tangent Problem for various functions.

We say a line is tangent to an object if it touches it only once.



The Tangent Problem Given a function f and a point c , determine the equation of the line tangent to f at the point $(c, f(c))$.



We can approximate the solution using a secant line. The slope of the secant line (difference quotient)

$$\text{Slope} = \frac{y_2 - y_1}{x_2 - x_1} = \frac{f(c+h) - f(c)}{(c+h) - c} = \frac{f(c+h) - f(c)}{h} \approx \text{slope of the tangent line}$$

NOTE: We get a better answer by moving closer to c (make h smaller).

Def The derivative of f at $x=c$ is defined by

$$f'(c) \stackrel{\text{def}}{=} \lim_{h \rightarrow 0} \frac{f(c+h) - f(c)}{h}$$

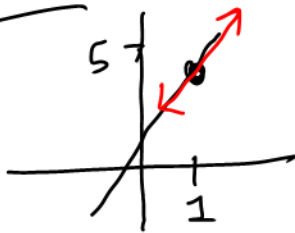
" f prime of c "

If $f'(c)$ exists, we say f is differentiable at c .

NOTE: (1) Some textbooks use Δx as opposed to h .

② If $f'(c)$ exists, then the line with slope $f'(c)$ containing the point $(c, f(c))$
 $y - f(c) = f'(c)(x - c)$
 is the solution to the Tangent Problem.

Ex 1 Find $f'(1)$ if $f(x) = 2x + 3$



$$f'(1) = \lim_{h \rightarrow 0} \frac{f(1+h) - f(1)}{h} = \lim_{h \rightarrow 0} \frac{2(1+h) + 3 - [2(1) + 3]}{h}$$

$$= \lim_{h \rightarrow 0} \frac{2 + 2h + 3 - 2 - 3}{h} = \lim_{h \rightarrow 0} \frac{2h}{h} = 2$$

Derivative As A Function

A point x $\rightarrow$ $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$ $\rightarrow$ The slope of the tangent line at the point x

Ex 2 If $f(x) = x^2$, find $f'(x)$

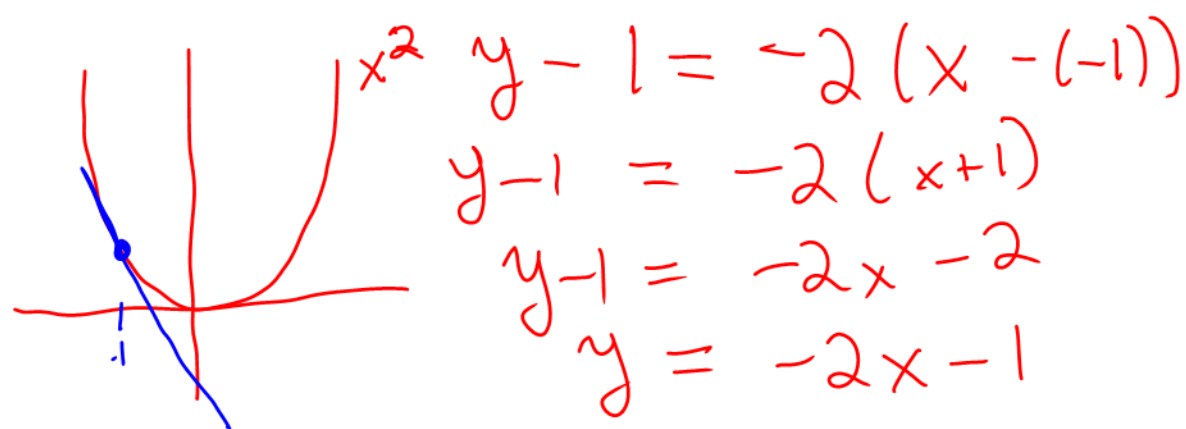
$$f'(x) = \lim_{h \rightarrow 0} \frac{(x+h)^2 - x^2}{h} = \lim_{h \rightarrow 0} \frac{x^2 + 2xh + h^2 - x^2}{h}$$

$$= \lim_{h \rightarrow 0} \frac{2xh + h^2}{h} = \lim_{h \rightarrow 0} [2x + h] = 2x$$

Ex 2b Find the equation of the line tangent to x^2 at $x = -1$

Slope: $f'(-1) = 2(-1) = -2$

Point: $(-1, f(-1)) = (-1, (-1)^2) = (-1, 1)$



NOTATION: $y = f(x)$

Derivative

$$f'(x)$$

$$y'$$

$$\frac{df}{dx}$$

$$\frac{dy}{dx}$$

Evaluate at c

$$f'(c)$$

$$y'(c) = y' \Big|_{x=c}$$

$$\frac{df}{dx} \Big|_{x=c}$$

$$\frac{dy}{dx} \Big|_{x=c}$$

Ex 3 $\frac{d}{dx}(x^3)$ Google Pascal's Triangle

$$\frac{d}{dx}(x^3) = \lim_{h \rightarrow 0} \frac{(x+h)^3 - x^3}{h} = \lim_{h \rightarrow 0} \frac{x^3 + 3x^2h + 3xh^2 + h^3 - x^3}{h}$$

$$= \lim_{h \rightarrow 0} \frac{3x^2h + 3xh^2 + h^3}{h}$$

$$= \lim_{h \rightarrow 0} [3x^2 + 3xh + h^2] = 3x^2$$

Example 4 Find y' if $y = \sqrt{x}$ ($x > 0$) $[a^2 - b^2 = (a-b)(a+b)]$

$$y' = \lim_{h \rightarrow 0} \frac{\sqrt{x+h} - \sqrt{x}}{h} \left(\frac{\sqrt{x+h} + \sqrt{x}}{\sqrt{x+h} + \sqrt{x}} \right)$$

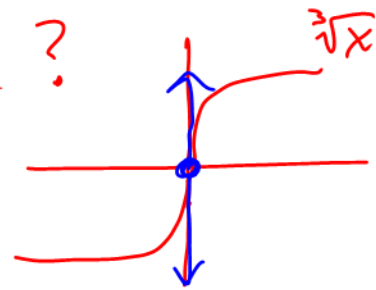
$$\begin{aligned}
 &= \lim_{h \rightarrow 0} \frac{(x+h) - (x)}{h[\sqrt{x+h} + \sqrt{x}]} \quad \frac{\frac{\sqrt{2}}{2}}{\frac{1}{\sqrt{2}}} \\
 &= \lim_{h \rightarrow 0} \frac{h}{h[\sqrt{x+h} + \sqrt{x}]} = \lim_{h \rightarrow 0} \frac{1}{\sqrt{x+h} + \sqrt{x}} \\
 &= \frac{1}{2\sqrt{x}}
 \end{aligned}$$

Question: When does differentiability fail?

Answer 1 A vertical tangent line

Example 5 When is $\sqrt[3]{x}$ differentiable?

Take for granted $\frac{d}{dx}(\sqrt[3]{x}) = \frac{1}{3\sqrt[3]{x^2}}$



Answer 2 Discontinuity

Theorem ① If f is differentiable at c , f is continuous at c .

② If f is discontinuous at c , then it is not differentiable at c .

Example 6 $\frac{d}{dx}(\frac{1}{x})$

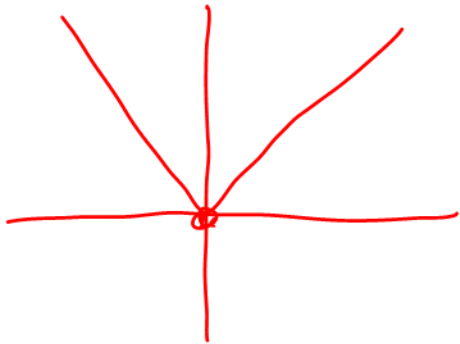
$$\lim_{h \rightarrow 0} \frac{\frac{1}{x+h} - \frac{1}{x}}{h} = \lim_{h \rightarrow 0} \frac{1}{h} \left[\frac{x}{(x+h)x} - \frac{x+h}{x(x+h)} \right]$$

$$= \lim_{h \rightarrow 0} \frac{1}{h} \left[\frac{x - x - h}{x(x+h)} \right] = \lim_{h \rightarrow 0} \frac{1}{h} \left[\frac{-h}{x(x+h)} \right] = -\frac{1}{x^2}$$

$$\Rightarrow \frac{d}{dx} \left(\frac{1}{x} \right) = \begin{cases} -\frac{1}{x^2} & x \neq 0 \\ \text{undefined} & x = 0 \end{cases}$$

Answer 3 Any kind of corner or kink

Ex 7 $|x|$ is not differentiable at 0.



This is an example of a function that is continuous at 0, but not differentiable at 0.